

2026 Spring

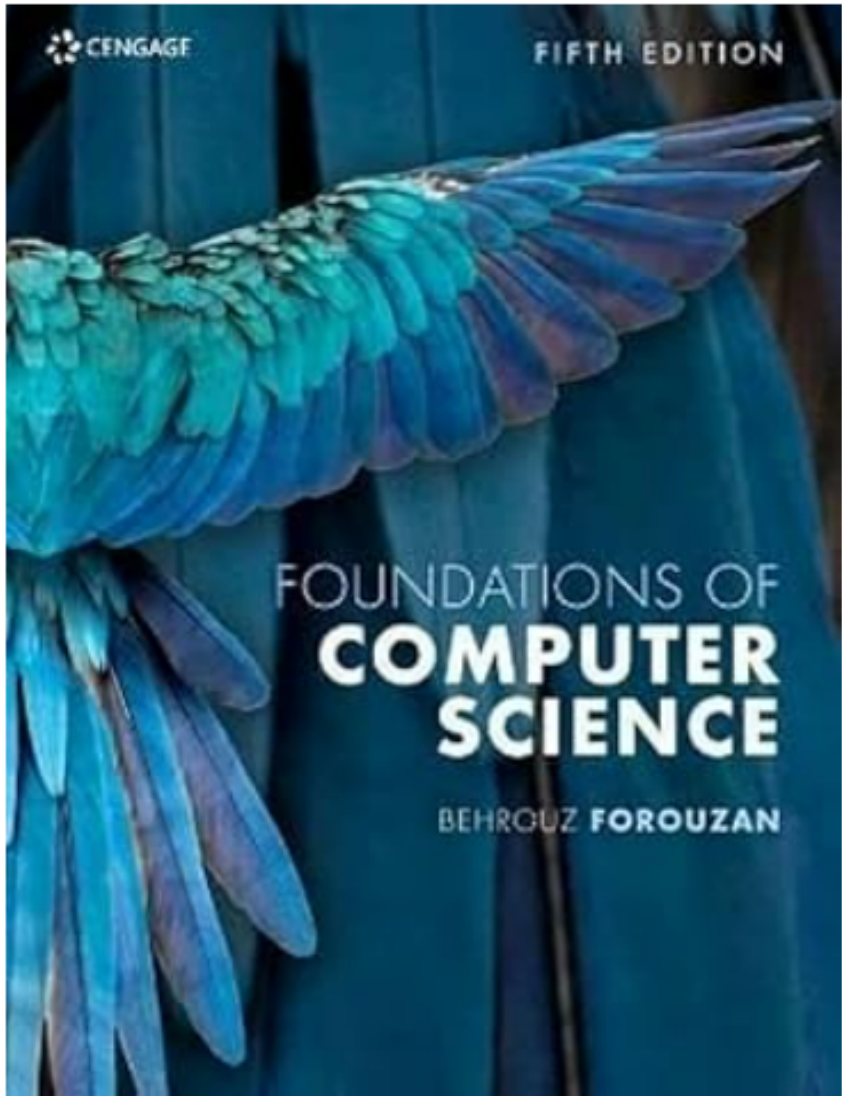
Course ID: 1142-1764

Foundations of Computer Science



12-Databases

The Textbook



Foundations of Computer Science Paperback



作者 [Behrouz Forouzan](#) (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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Flip Classroom

Padlet Collaborative Learning

Introduction to Computer Science | Modules 5 – 10



M5 Networking



M6 Compute



M7 Storage



M8 Databases



M9 Architecture



M10 Auto Scaling



Your Group Builds the Knowledge Base — Together

Grading

The report's grading includes:

- 20% teacher's score
- 20% group score
- 60% peer evaluation.

Group leaders:

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What Is a Flip Classroom?

1 BEFORE Class

Study the module
watch videos,
read materials at home



2 PREPARE Padlet

Your group fills in
all 5 sections of
the Padlet template



3 IN Class

Present & discuss
answer classmates'
questions (Q&A)



4 REFLECT

Respond to peer
comments, update
the Padlet



Padlet is the shared workspace where your group documents your learning for each module — visible to the whole class!



Your Padlet Template — 5 Sections

S1



Learning Objectives

State the key learning goals of the module.
What should you understand by the end?

S2



Core Concepts

Summarize essential technical concepts,
terminology, and architectures.

S3



Labs

Document hands-on AWS lab steps,
screenshots, and observations.

S4



Challenges

Record difficult questions, errors
you encountered, and how you solved them.

S5



Q&A & Peer Support

Post questions for the class.
Answer and support your peers.

 **Tips for a Great Padlet**
Add images & diagrams, not just text

Link to AWS documentation & tutorials

Use clear headings in each column

Respond to at least 2 peers in S5



AWS Academy

Cloud Foundations

Module 8

Databases

Overview of AWS managed database services —
Relational, NoSQL, Data Warehousing, Caching & Migration

Learning Objectives

By the end of this module, you will be able to:

- ✓ Describe the AWS managed database services
- ✓ Explain Amazon RDS features, use cases, and supported engines
- ✓ Describe Amazon Aurora and its advantages
- ✓ Understand Amazon DynamoDB as a NoSQL key-value store
- ✓ Differentiate Amazon Redshift for data warehousing
- ✓ Explain Amazon ElastiCache for in-memory caching
- ✓ Describe the AWS Database Migration Service (DMS)

Why Use AWS Managed Databases?



No Undifferentiated Heavy Lifting

AWS handles hardware provisioning, patching, backups, and failure detection automatically.



High Availability & Durability

Multi-AZ deployments and automatic failover ensure minimal downtime.



Security Built-In

Data encryption at rest and in transit, VPC isolation, and IAM-based access control.



Pay-as-You-Go Pricing

No upfront costs. Scale compute and storage independently; pay only for what you use.

AWS Database Services at a Glance

Service	Category	Key Characteristics
Amazon RDS	Relational	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Aurora
Amazon Aurora	Relational	MySQL/PostgreSQL-compatible; up to 5× faster
Amazon DynamoDB	Key-Value/Doc	Single-digit millisecond performance at any scale
Amazon Redshift	Data Warehouse	Petabyte-scale analytics with columnar storage
Amazon ElastiCache	In-Memory	Redis & Memcached; sub-millisecond latency
Amazon Neptune	Graph	Fully managed graph database for connected data

Amazon RDS — Relational Database Service

What is Amazon RDS?

Amazon RDS is a managed relational database service that makes it easy to set up, operate, and scale relational databases in the cloud.

AWS manages:

Hardware provisioning

Database setup

Patching

Backups

Monitoring & alerting

Supported Database Engines

MySQL

Open-source; widely used

PostgreSQL

Advanced open-source RDBMS

MariaDB

Community-driven MySQL fork

Oracle

Enterprise-grade commercial DB

SQL Server

Microsoft's enterprise RDBMS

Amazon Aurora

AWS-native, MySQL/PG compat.

Amazon RDS — High Availability & Scalability

Multi-AZ Deployments

- Synchronously replicates data to a standby instance in a different Availability Zone
- Automatic failover to standby if primary fails (DNS endpoint unchanged)
- Designed for high availability — NOT for read scaling
- Failover typically completes in 1–2 minutes

Read Replicas

- Asynchronously replicated copies of the primary instance
- Offload read-heavy workloads (analytics, reporting) from primary
- Can be promoted to a standalone DB if needed
- Available for MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, and Aurora

Amazon Aurora — Cloud-Native Relational Database

Amazon Aurora is a MySQL- and PostgreSQL-compatible relational database built for the cloud, combining the performance and availability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases.

5x

Faster than MySQL

3x

Faster than PostgreSQL

15

Replicas per cluster

99.99%

Availability SLA

Distributed Storage

Data is automatically split across 100s of storage nodes across 3 AZs; 6 copies of data maintained.

Aurora Serverless

On-demand auto-scaling config; the DB automatically starts, shuts down, and scales based on application demand.

Global Database

One primary region with up to 5 secondary regions; typical cross-region replication latency under 1 second.

Continuous Backup

Automated backup to S3 with sub-5-second granularity; no performance impact.

Amazon DynamoDB — NoSQL Key-Value Database



Amazon DynamoDB

Fully managed NoSQL

Key-value & document store

Single-digit ms performance

Serverless — no cluster management

Auto scales to handle load

Key Features & Concepts

Tables, Items & Attributes

No fixed schema. Each item can have different attributes. Partition key (+ optional sort key) is required.

On-Demand & Provisioned Capacity

On-demand mode auto-scales. Provisioned mode lets you set read/write capacity units (RCU/WCU).

DynamoDB Streams

Captures time-ordered sequence of item-level changes; trigger Lambda functions for event-driven architectures.

Global Tables

Multi-Region, multi-active replication — < 1s replication; ideal for globally distributed apps.

Amazon Redshift — Cloud Data Warehouse

Petabyte-scale, fully managed data warehouse for analytics and business intelligence workloads

Architecture

Massively Parallel Processing (MPP)

Leader node + compute nodes

Columnar data storage

Result caching for repeated queries

Performance

10× faster than traditional data warehouses

Advanced compression reduces storage

Redshift Spectrum: query S3 directly

Concurrency Scaling handles bursts

Use Cases

Business intelligence & reporting

Log analysis and clickstream data

Financial analytics

Marketing attribution analysis



Redshift is optimized for OLAP (analytics) — not for OLTP (transactional) workloads. Use RDS or Aurora for transactional databases.

Amazon ElastiCache — In-Memory Caching

Redis

Rich data structures (lists, sets, sorted sets, hashes)

Persistence options (snapshots + AOF)

Multi-AZ replication with automatic failover

Pub/Sub messaging, Lua scripting

Cluster mode: horizontal sharding up to 500 nodes

Memcached

Simple key-value store (strings only)

Multi-threaded architecture; good CPU utilization

No persistence or replication

Auto Discovery: clients find new/removed nodes

Best for simple, stateless caching needs

Use Case: Reduce database load by caching frequently accessed query results. Sub-millisecond response times.

AWS Database Migration Service (DMS)

Migrate databases to AWS quickly and securely with minimal downtime. The source database remains operational during migration.



Homogeneous Migration

Same engine (e.g. MySQL → RDS for MySQL). Schema is largely compatible; AWS Schema Conversion Tool (SCT) not required.

Heterogeneous Migration

Different engines (e.g. Oracle → Aurora PostgreSQL). Use AWS SCT to convert schema and code first, then DMS for data.

Continuous Replication

Change Data Capture (CDC) keeps source and target in sync during migration; cut over with minimal downtime.

Supported Sources & Targets

Oracle, SQL Server, MySQL, MariaDB, PostgreSQL, SAP, MongoDB, S3, and more — to RDS, Aurora, DynamoDB, Redshift, S3.

Choosing the Right AWS Database Service

Service	Workload Type	Primary Use Case	Scaling Model
Amazon RDS / Aurora	Transactional (OLTP)	E-commerce, ERP, CMS, SaaS apps needing relational SQL	Vertical + Read Replicas
Amazon DynamoDB	Key-Value / Document	Gaming leaderboards, IoT data, session stores, real-time	Seamless horizontal
Amazon Redshift	Analytical (OLAP)	Business intelligence, data lakes, big data analytics	MPP cluster scaling
Amazon ElastiCache	In-Memory Cache	Session cache, query cache, leaderboards, real-time ads	Cluster with sharding
Amazon Neptune	Graph Database	Social networks, fraud detection, knowledge graphs	Read replicas

Module 8 Summary

Key Takeaways

Amazon RDS

Managed relational DB for MySQL, PostgreSQL, MariaDB, Oracle, SQL Server. Multi-AZ for HA; Read Replicas for scale.

Amazon Redshift

MPP columnar data warehouse for OLAP; petabyte-scale; Redshift Spectrum to query S3 without loading data.

Amazon Aurora

Cloud-native RDBMS; 5× MySQL speed, 3× PostgreSQL speed; 6 copies across 3 AZs; Aurora Serverless for variable workloads.

Amazon ElastiCache

In-memory caching with Redis or Memcached; sub-ms latency; offloads databases from repetitive read queries.

Amazon DynamoDB

Fully managed NoSQL; single-digit ms at any scale; serverless; Global Tables for multi-region active-active replication.

AWS DMS

Migrate databases with minimal downtime; supports homogeneous & heterogeneous migrations; CDC for continuous replication.